

Class- 11 Interrupt Class Program

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/*
 * File:  main.c
 * Author:  sagar
 *
 * Created on 10 June, 2025, 9:46 AM
 */

// CONFIG
#pragma config FOSC = HS      // Oscillator Selection bits (HS oscillator)
#pragma config WDTE = OFF     // Watchdog Timer Enable bit (WDT disabled)
#pragma config PWRTE = OFF    // Power-up Timer Enable bit (PWRT disabled)
#pragma config BOREN = OFF    // Brown-out Reset Enable bit (BOR disabled)
#pragma config LVP = OFF      // Low-Voltage (Single-Supply) In-Circuit Serial Programming
                             // Enable bit (RB3 is digital I/O, HV on MCLR must be used for programming)
#pragma config CPD = OFF      // Data EEPROM Memory Code Protection bit (Data EEPROM
                             // code protection off)
#pragma config WRT = OFF      // Flash Program Memory Write Enable bits (Write protection off;
                             // all program memory may be written to by EECON control)
#pragma config CP = OFF       // Flash Program Memory Code Protection bit (Code protection off)

#include <xc.h>
#define _XTAL_FREQ 20000000

unsigned char value;

void main(void)
{
    TRISB = 0xF0; // RB4 to RB7 Input
    TRISC = 0x00; // PORTC as Output
    TRISD = 0x00; // PORTD as Output

    PORTB = 0x00;
    PORTC = 0x00;
    PORTD = 0x00;

    OPTION_REG &= ~(0x01 << 7); // PORTB internal pull up

    INTCON = 0x88; // Enable interrupt

    while(1)
    {
        /* Thread Mode */
        PORTD = 0xFF;
        __delay_ms(1000);

        PORTD = 0x00;
        __delay_ms(1000);
    }
}
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    }
    return;
}

/* Interrupt Flag bit */
void __interrupt() _ISR_Function()
{
    if(INTCON & 0x01)
    {
        value = PORTB;

        if(value == 0xE0)
        {
            PORTC = ~PORTC;
        }

        /* Clear the Flag */
        INTCON &= ~(0x01 << 0);
    }
}

```